

4(a)	(when two or more) waves meet / overlap (at a point)	B1
	(resultant) displacement is the sum of the individual displacements	B1
4(b)	$\lambda = ax / D$	C1
	$\lambda = [(0.65 \times 10^{-3}) \times (1.7 \times 10^{-3})] / 1.9$	C1
	$\lambda = 5.8 \times 10^{-7} \text{ m}$	A1
4(c)(i)	waves are out of phase by a quarter of a cycle / period or when one wave has maximum/minimum displacement, the other wave has zero displacement	B1
4(c)(ii)	(statement 2 is) not correct (because) waves do not have phase difference of 180° / not in antiphase or one wave has some displacement when other has no displacement or the displacements of the waves are not always equal and opposite	B1
4(d)	more diffraction (of light/waves by slits) or light/waves are more spread (by slits) or light/waves (from slits) have less intensity	B1
	more (bright/dark) fringes or bright fringes are less bright / are dimmer / have lower intensity or no change to fringe spacing/separation/width	B1